

X-ray Signature of Colliding Wind Wolf-Rayet Stars with a Supernova Event

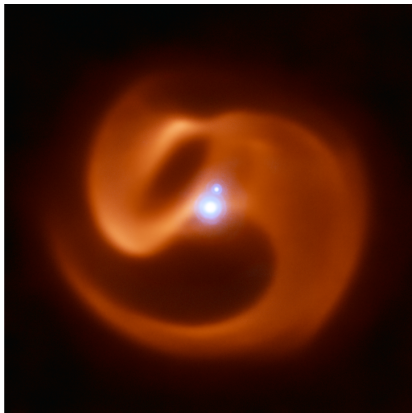
Sam, Vincent, Divyansh

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Introduction

- **The Apep System:** A complex triple stellar system located in the Milky Way consisting of:
 - A central binary system of two Wolf-Rayet (WR) stars of subtypes WC8 and WN4-6b.
 - An O8 Iaf supergiant positioned approximately 1700 AU to the north.
- This system represents one of the strongest cases of a classical WR+WR binary system in our galaxy.
- Dynamics include wind velocities of 2100 ± 200 km/s for the WC8 star and 3500 ± 100 km/s for the WN4-6b star.
- Thinking of adding a supernova event to the system [O8 Iaf].

Apep Star System: Cosmic Serpent



Initial Conditions

- **Binary WR Stars (WC8 and WN4-6b):**

- Period: ~ 100 years
- Masses: $\sim 18 M_{\odot}$ (WC8) and $\sim 15 M_{\odot}$ (WN4-6b)
- Wind velocities: WC8: 2100 ± 200 km/s, WN4-6b: 3500 ± 100 km/s.
- Mass-loss rates: WC8: $10^{-4.5} M_{\odot} \text{yr}^{-1}$, WN4-6b: $10^{-4.3} M_{\odot} \text{yr}^{-1}$.
- Expected momentum ratio, η : ~ 0.4 .

- **O8 Iaf Supergiant:**

- Distance from binary: Approximately 1700 AU
- Mass: Typical values for O8 Iaf supergiants.

AMUSE Tools and Codes

- **Stellar Evolution:**

- **SEBA:** Will use for the evolution of the individual stars in the system. Mainly to provide stellar evolution tracks, allowing us to model changes in the stellar parameters over time.

- **Binary System Dynamics:**

- **N-body code, ph4:** For the gravitational interactions between the stars in the Apep system. Mainly for the orbital evolution of the stars.

- **Important! Hydrodynamics:**

- **Fi, Flash:** Used to simulate the wind-wind collision between the WR stars and any later interactions with the supergiant's wind. Modeling the shocks and any other hydrodynamic phenomena.

- **Radiative Transfer:**

- **Radiative transfer code (to be determined):** To make the X-ray signatures generated by the colliding winds and the potential supernova event which can be compared with actual observations (as suggested by Simon)

Modeling the Supernova Event

- Evolving the star for some time and then the supernova of the O8 Iaf supergiant.
- Potential effects on the WR binary due to the supernova: only radiative effects? [maybe gravitational or material effects].

Pseudo Code

INITIALIZE AMUSE framework

```
// Load Stellar Evolution, Hydrodynamics, Radiative Transfer, and n-body code
IMPORT SEBA, MESA/HydroCode, RadiativeTransfer, ph4
```

```
// Define initial conditions
```

```
DEFINE WC8 star with:
```

- Mass = 18 solar Mass
- Position = value such that binary separation is stable

```
DEFINE WN4-6b star with:
```

- Mass = 15 solar mass
- Position = value such that binary separation is stable and period is around 100 years

```
DEFINE O8 Iaf supergiant with:
```

- Mass = around 20 solar mass
- Position = 1700 au from binary

```
// Combine stars into a single stellar system
```

```
CREATE Apep_system from WC8, WN4-6b, and O8 Iaf
```

```
// Initialize ph4 for gravitational interactions
```

```
INITIALIZE ph4_gravity = ph4()
```

```
// Add Apep_system to ph4 gravity solver
```

```
ADD Apep_system to ph4_gravity
```

```
// Define simulation parameters
```

```
SET timestep = [small time interval]
```

```
SET total_simulation_time = [desired simulation length] # some time before supernova and then after
```

Pseudo Code

```
WHILE current_time < total_simulation_time:

    // Calculate gravitational interactions using ph4
    CALCULATE gravitational_forces using ph4_gravity(Apep_system)

    // Update stellar evolution using SEBA
    EVOLVE stars using SEBA(Apep_system, timestep)

    // Calculate wind interactions using the hydrodynamics code
    CALCULATE wind_interactions using HydroCode(Apep_system)

    // Calculate the X-ray flux signature using radiative transfer
    CALCULATE xray_flux using RadiativeTransfer(Apep_system)

    // Update positions and velocities based on gravitational interactions
    UPDATE positions and velocities(Apep_system, timestep)

    // Check for potential supernova event
    IF 08 Iaf meets supernova criteria:
        //Inject supernova energy
        // Calculate the X-ray flux signature after supernova using radiative transfer
        CALCULATE post_supernova_xray_flux using RadiativeTransfer(Apep_system)
        BREAK and ALERT "Supergiant is going supernova!"

    // Save or visualize the current state if desired
    IF current_time % [some interval] == 0:
        SAVE or DISPLAY state(Apep_system)

    // Increment the time
    current_time += timestep

END
```

What we want to produce in the Results

X-ray Signature of Colliding Winds

- Try to reproduce the serpent-like figure.
- Displaying X-ray signatures from WR binary winds collision before the supernova.
- Factors affecting the signatures: momentum ratio, wind velocities, and shocks.

X-ray Signature Post-Supernova

- Changes in the X-ray signature after the supernova.
- New material injected, impact on wind collisions, and overall system dynamics.

Geometric model of colliding-wind dust plume

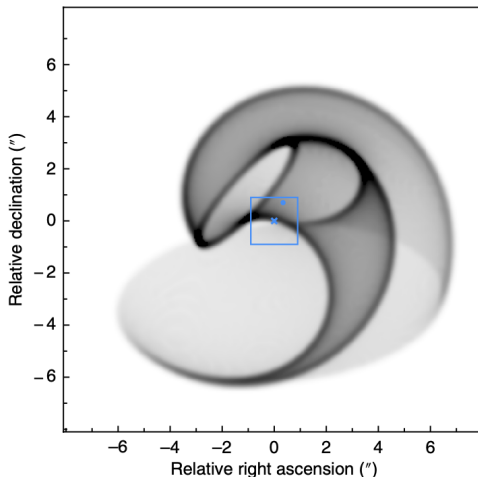


Fig. 4 | A geometric model of the dust plume of Apep. Note that the model only traces the surface of the colliding-wind dust plume. The mathematical origin of the Archimedian spiral (and the presumed location of the stars of the Central Engine) is indicated by the blue cross centred within the box.

What we want to produce in the Results

Try to make a plot and then look for the same in the catalogues.

